INCH-POUND

MIL-T-47012A(MI)
28 June 1989
SUPERSEDING
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19 April 1974

MILITARY SPECIFICATION

TAPE, PRESSURE-SENSITIVE ADHESIVE, COPPER FOIL

This specification is approved for use within the Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification defines one type of copper foil electrical tape with an electrically conductive pressure-sensitive adhesive.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-B-585	-	Boxes, Wood, Wirebound
PPP-B-601	-	Boxes, Wood, Cleated-Plywood
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	-	Boxes, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5270 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A
FSC 8030
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MILITARY

MIL-P-116 - Preservation, Methods of

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts
MIL-STD-1190 - Minimum Guidelines for Level C Preservation and Packaging

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 1000 - Pressure-Sensitive Adhesive Coated Tapes Used for Electrical Insulation, Standard Methods of Testing

(Application for copies should be addressed to the American Society for Testing and Materials, 1016 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to a first article inspection (see 4.4 and 6.3).

3.2 Material. The tape shall consist of 0.0014 inch thick (1 ounce) copper foil, coated on one side with an electrically conductive pressure-sensitive adhesive. Unless otherwise specified (see 6.2), the length and width shall be as specified in the contract or order.

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3.3 Mechanical properties.

3.3.1 Tensile strength. The tensile strength of the tape shall be not less than 25 pounds per inch.

3.3.2 Adhesion.

3.3.2.1 Adhesion after room temperature exposure. After exposure to ambient room temperature for 15 to 25 minutes, the tape shall have an adhesion level of 30 ounces per inch.

3.3.2.2 Adhesion after exposure to high temperatures. The tape shall be applied to a suitable substrate and aged at 130 ± 3° Celsius (C) (266 ± 10° Fahrenheit [F]) for 55 to 65 hours. The aged tape shall have an adhesion level of not less than 45 ounces per inch of width, measured at ambient room temperature.

3.4 Physical properties.

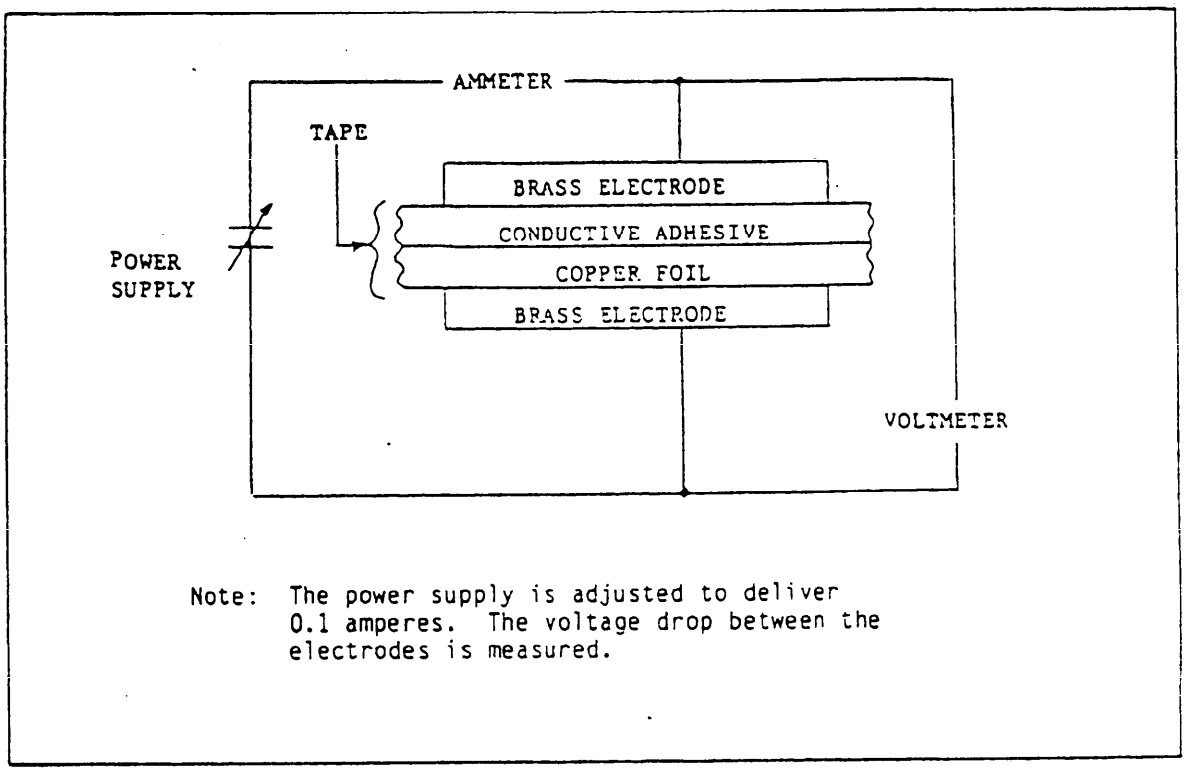
3.4.1 Thickness. The total thickness of the tape shall be 0.0035 ± 0.0005 inch.

3.4.2 Separator. A separator or liner shall be interposed between adjacent layers of tape within the roll.

3.5 Electrical requirements.

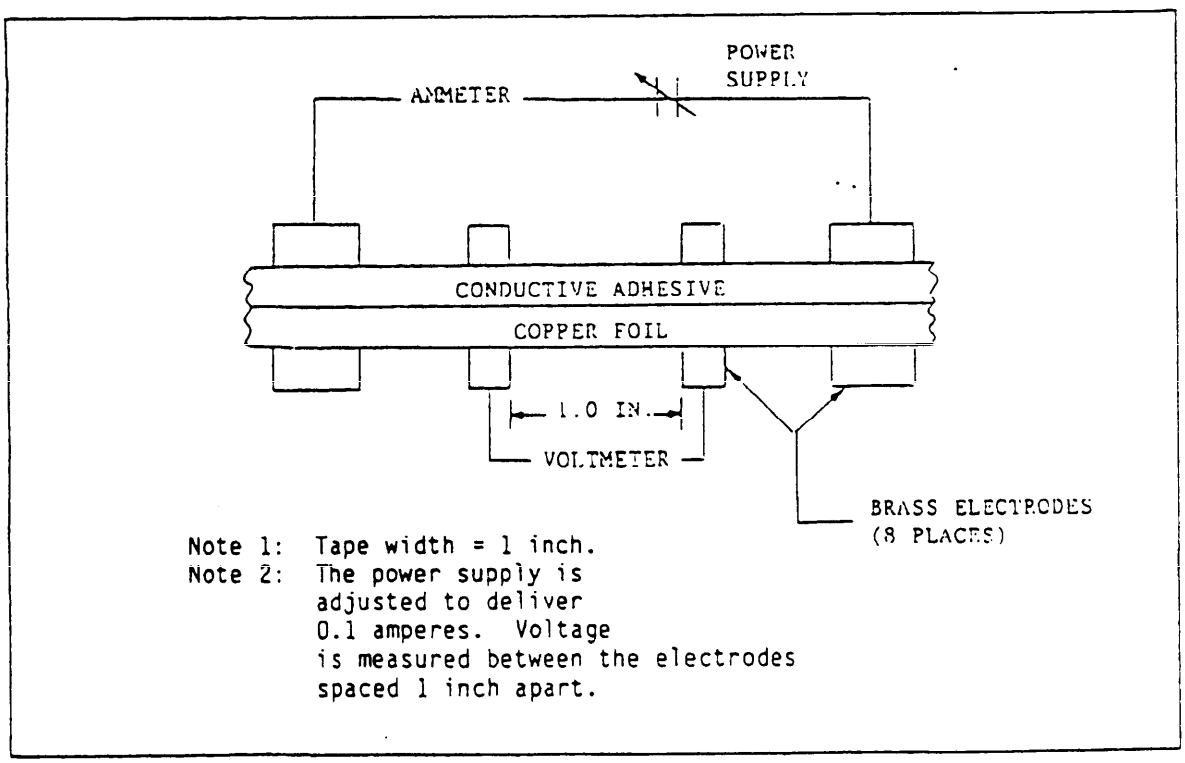
3.5.1 Resistance. Using a contact pressure of 5 pounds per square inch (psi), resistance shall be measured in accordance with figures 1 and 2, by substituting values in the formula $R = \frac{V}{I}$, where R is resistance, V is voltage, and I is current in amperes. Resistance through the tape shall be 0.01 ohms per square inch (maximum) when tested in accordance with figure 1 and using 1 square inch brass electrodes. Resistance along the length of the tape shall be 0.0065 ohms per square inch (maximum) when tested in accordance with figure 2.

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Note: The power supply is adjusted to deliver 0.1 amperes. The voltage drop between the electrodes is measured.

FIGURE 1. Resistance measured through tape.



Note 1: Tape width = 1 inch.
Note 2: The power supply is adjusted to deliver 0.1 amperes. Voltage is measured between the electrodes spaced 1 inch apart.

FIGURE 2. Resistance measured along length of tape.

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3.6 Shelf life. The shelf life of the tape in the packaged condition shall be not less than one year from the date of manufacture, when stored at temperatures less than +27°C (+80.6°F).

3.7 Workmanship. The workmanship of the tape shall ensure a high quality product that is uniform and in conformance with the requirements of this specification. The copper foil tape finish shall be smooth, uniform, free from dirt, lumps, or bare spots, and shall be free of foreign material or other contaminants that may affect appearance or serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)

b. Quality conformance inspection (see 4.5)

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.6, 4.6.1 and 4.7. Preproduction inspection shall be performed by the Government at a place specified by the procuring activity (see 6.2).

4.4 First article inspection. First article inspection shall be conducted on the first article sample (see 3.1) and also as specified by the procuring activity (see 6.2). If a lot should fail a first article inspection, no further lot will be accepted until the supplier has presented sufficient evidence to show that the condition which caused the failure has been corrected. The first article inspection shall consist of all the tests of this specification (see 4.5 and table I).

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4.5 Quality conformance inspection. Quality conformance inspections shall be specified in table I. The inspection of commercial packaging shall be as specified in the contract (see 6.2).

TABLE I. Quality conformance inspection.

Inspection	Requirement Paragraph	Method Paragraph
Material	3.2	Visual
Tensile strength	3.3.1	4.7.1.1
Adhesion after room temperature exposure	3.3.2.1	4.7.1.2
Adhesion after high temperature exposure	3.3.2.2	4.7.1.2
Thickness	3.4.1	4.7.2.1, 4.7.2.2
Separator	3.4.2	Visual
Resistance	3.5.1	4.7.3
Shelf life	3.6	4.7.4
Workmanship	3.7	4.8

4.6 Sampling for conformance.

4.6.1 Lot formation. The term "lot", as used in this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, specification, or revision thereof. Changes to either the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government inspector, shall not be deemed to alter the homogeneity of the lot. Lot size shall consist of all the tape manufactured at one time, from one batch, forming parts of one contract or order, and submitted at the same time and place.

4.6.2 Sampling. Unless otherwise specified (see 6.2), tape representative of each lot shall be selected at random for quality conformance testing. Each roll of tape shall be considered as a unit of product. Failure of the sample to meet the quality conformance requirements shall be cause for lot rejection.

4.7 Test methods.

4.7.1 Mechanical properties.

4.7.1.1 Tensile strength. The tensile strength shall be determined in accordance with ASTM D 1000. The tensile strength shall be as specified in 3.3.1.

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4.7.1.2 Adhesion. Adhesion shall be determined in accordance with ASTM D 1000, using test specimens measuring 10.0 ± 0.126 inches long and 1.0 ± 0.032 inches wide. Adhesion levels shall be as specified in 3.3.2.1 and 3.3.2.2.

4.7.2 Physical properties.

4.7.2.1 Total thickness. The thickness of the tape, including foil and adhesive, shall be determined in accordance with ASTM D 1000. Total thickness shall be as specified in 3.4.1.

4.7.2.2 Foil thickness. The thickness of the foil shall be determined in accordance with ASTM D 1000. Total foil thickness shall be as specified in 3.2.

4.7.3 Resistance. Resistance shall be determined in accordance with MIL-STD-202, Method 303, using the test configuration shown in figures 1 and 2. Resistance shall be as specified in 3.5.1.

4.7.4 Shelf life. The tape shall be tested for conformance with all requirements specified herein after completion of the storage period specified in 3.6. A manufacturer's certificate of compliance with the shelf life requirement may be accepted during the storage period.

4.8 Workmanship. Workmanship of the finished product shall be visually examined to determine compliance with 3.7.

5. PACKAGING

5.1 Preservation, packaging, unitization and marking. Preservation, packaging, unitization and marking shall be either level A, level B, or Industrial (see 6.2).

5.1.1 Level A. Tape shall be packaged in accordance with MIL-P-116, Method III, using a container conforming to PPP-B-636, class weather-resistant.

5.1.2 Level B. Level B shall be the same as level A, except a domestic-type container shall be used.

5.1.3 Level C. Preservation and packaging shall be in accordance with MIL-STD-1190.

5.2 Packing.

5.2.1 Level A. The packaged tape shall be packed in containers conforming to PPP-B-585, Class 3, Style 2 or 3; PPP-B-601, overseas type; or PPP-B-621, Class 2, Style 4.

5.2.2 Level B. The packaged tape shall be packed in fiberboard containers conforming to PPP-B-636, class domestic, style optional.

5.2.3 Level C. Packing shall be in accordance with MIL-STD-1190.

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5.2.4 Unitization. When specified, packs shall be palletized in accordance with MIL-STD-147.

5.3 Marking.

5.3.1 Level A or level B. Marking shall be in accordance with MIL-STD-129.

5.3.2 Level C. Marking shall be in accordance with MIL-STD-1190.

5.3.3 Special marking. Shelf life markings may be required (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The copper foil tape described in this specification is designed for use in electrical or electronic applications where an electrically-conductive adhesive is desirable.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Whether a first article sample is required (see 3.1 and 4.4).

c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of industrial documents referenced (see 2.1.1 and 2.2).

d. Whether shelf life marking is required (see 5.3.3).

e. Location for preproduction inspections (see 4.3).

f. Sampling plans, if different than specified (see 4.3.2).

g. Requirements for preservation, packing, and packaging (see 4.5.1, 5.1 and 5.2).

h. Whether unitization is required (see 5.2.4).

i. Whether special markings are required (see 5.3.3).

j. Length and width required (see 3.2).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding

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arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Subject term (keyword) listing.

Adhesion
Coating, metallic
Bronze sheet, electrically-conductive
Cohesive
Copper film
Film, electrically-conductive
Filament
Pressure sensitive
Ribbon, metal coated

6.5 Metrication. Wherever inch/pound measurements are used in this document, metric equivalents in accordance with FED-STD-376 shall be acceptable.

6.6 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Project 8030-A123

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